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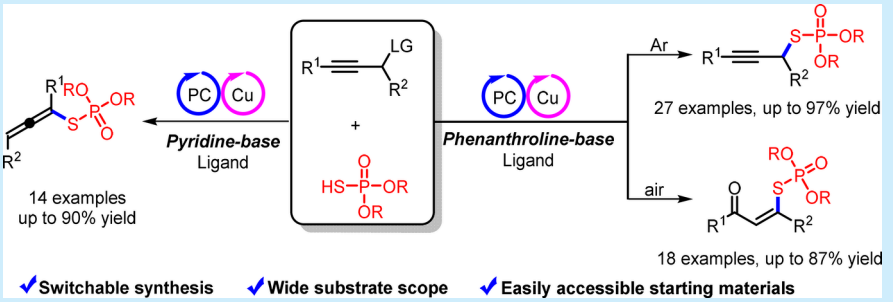
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Photoredox/Copper Dual-Catalyzed Phosphorothiolation of Propargylic Derivatives for the Switchable Synthesis of *S*-Alkyl, *S*-Vinyl and *S*-Allenyl Phosphorothioates

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ABSTRACT: Herein, we report a photoredox/copper dual-catalyzed selective phosphorothiolation of propargylic derivatives from easily accessible [P(O)SH] compounds. This reaction provides a general, mild and versatile procedure to synthesize a variety of synthetically useful *S*-alkyl, *S*-vinyl and *S*-allenyl phosphorothioates selectively from the same set of simple starting materials.

Phosphorothioates containing a phosphorus−sulfur single bond (P−S) are widely found in many pharmaceutical and agrochemical molecules (Scheme 1a).[1] Traditional methods, such as Michaelis−Arbuzov-type reaction, for the synthesis of phosphorothioates usually required thiols as the starting materials, which might suffer from high temperature, toxicity, and moisture sensitivity of reagents, thus resulting in harsh reaction conditions, complicated operation, and limited substrates (Scheme 1b, top).[2] By contrast, *S*-hydrogen phosphorothioates [P(O)SH][3−7] and their salts,[8] regardless of presynthesized or generated *in situ*, have been developed as an important synthon in phosphorothiolation processes under mild reaction conditions. As a result, this readily available, stable and low-toxicity [P(O)SH] reagent in phosphorothiolation reactions with olefins,[5] terminal alkynes[6] or via 1,5-hydrogen atom transfer (1,5-HAT) process[7] and so on[3,4,8] provide an important alternative for the synthesis of a variety of high-value phosphorothioates (Scheme 1b, bottom). However, the transformations involving [P(O)SH] reagents are still underdeveloped; therefore, achieving structurally diverse phosphorothioates from [P(O)SH] compounds remains a desirable yet challenging goal.

Propargylic derivatives, a kind of redox-active substrate, could be easily activated to generate propargyl species and its tautomeric allenyl species.[9,10] Benefiting from the extraordinary reaction characteristics of propargyl radicals, major advancements have been achieved in the transformation of propargyl derivatives into functionalized propargylic derivatives[9] or allenes[4,10] via metal-catalyzed coupling reactions. However, these elegant works are mostly limited to the intrinsic two-electron mechanism involving an oxidative addition-transmetalation-reductive elimination process, while examples involving a visible-light-induced through single electron transfer (SET) process are scarce.[9d,10j] In continuation of our efforts in copper-catalyzed phosphorothiolation with [P(O)SH] reagents,[3e,6b,7] we speculated that the reactive propargyl radicals and their tautomeric allenyl radicals could be generated from propargylic derivatives via an SET process which might be suitable for the construction of propargylic or allenylic phosphorothioates. On the other hand, we doubt that whether the highly reactive allenyl radicals could interact with oxygen to generate alkenyl radical[11] which might also be able to couple with [P(O)SH] reagents to produce vinylic phosphorothioates (Scheme 1c). The predictable challenges of this approach include (1) the regioselectivity issue on formation of aforementioned specific radicals and (2) the match of the specific radicals with the transition metal species

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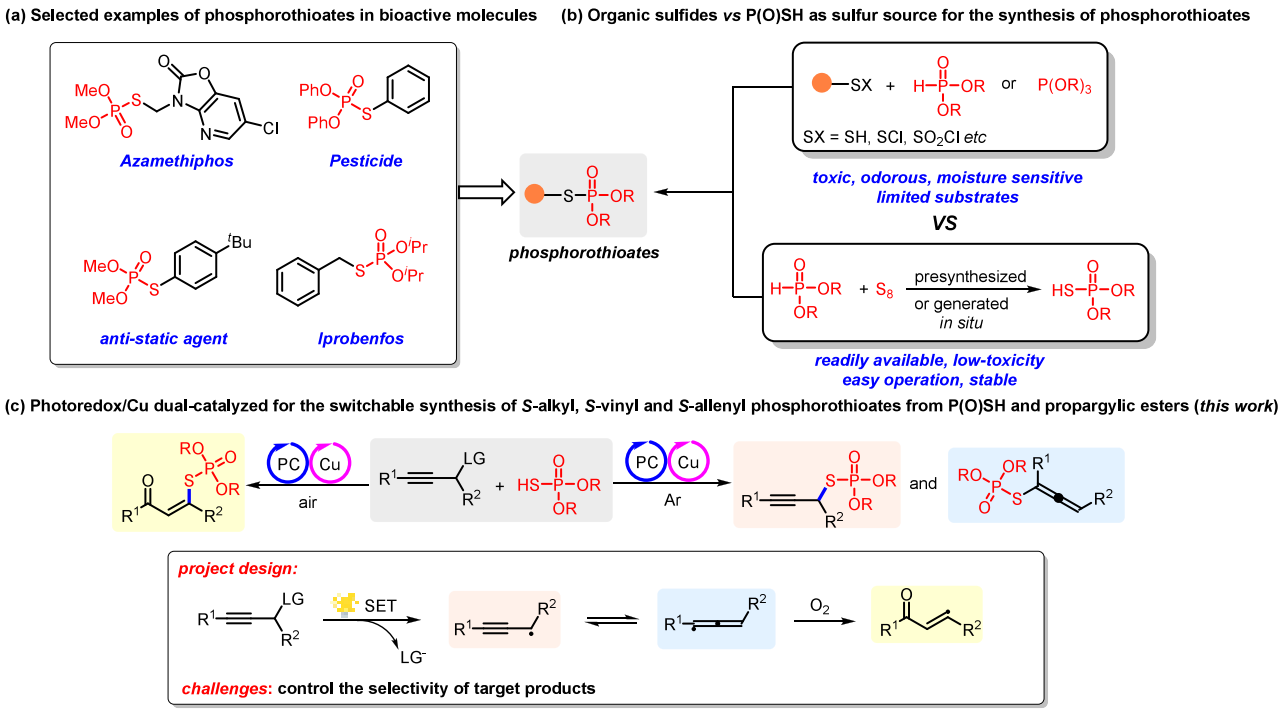


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Scheme 1. Application and Synthesis of Thiophosphates

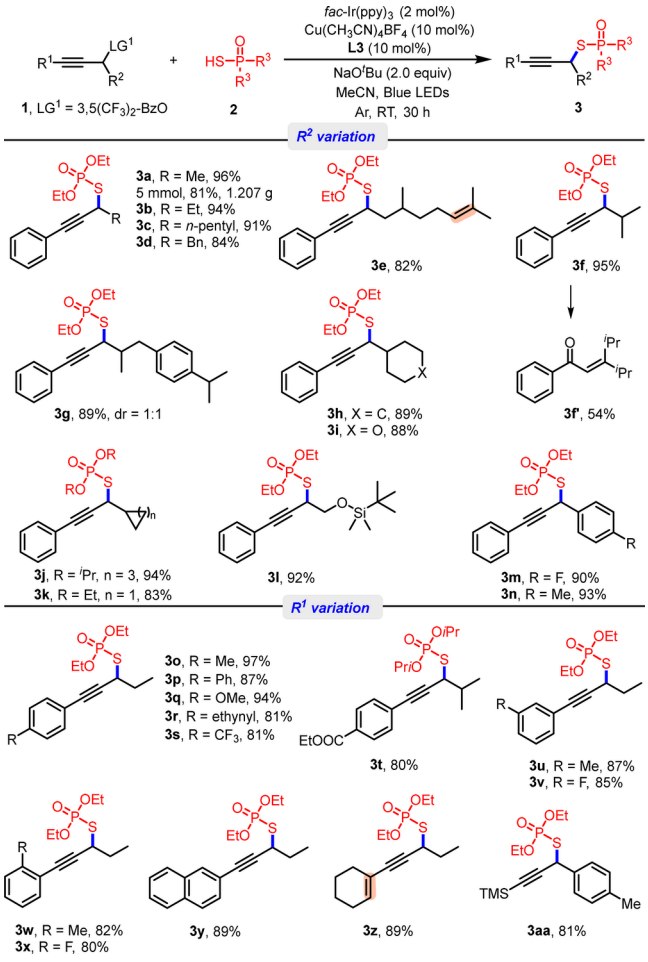


to yield target diverse products. Therefore, in this work, we targeted the development of a photoredox/copper dual-catalyzed phosphorothiolation of propargylic derivatives by employing [P(O)SH] compounds as coupling partners for the switchable synthesis of highly valued *S*-alkyl, *S*-vinyl and *S*-allenyl phosphorothioates (Scheme 1c).

To test our assumption, we started the reaction design by choosing propargyl ester bearing 3,5-$(F_3C)_2PhCO_2$- as the leaving group (**1a**) and $(EtO)_2P(O)SH$ (**2a**) as the model substrate irradiated by blue LEDs under an Ar atmosphere (for details see Table S1). The desired *S*-alkyl phosphorothioate **3a** was obtained in 96% yield in the presence of *fac*-Ir(ppy)$_3$, $Cu(MeCN)_4BF_4$, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (**L3**) and NaO^{t}Bu in MeCN at room temperature. The anticipated *S*-allenyl phosphorothioate was not detected under the current reaction conditions. Then the reaction was conducted under air to verify our assumption. Pleasingly, the (*Z*)-*S*-vinyl phosphorothioate **4a** was obtainted in 86% yield by using 4CzIPN as the photocatalysis, CuCl as the catalyst, 4,7-diphenyl-1,10-phenanthroline (**L1**) as the ligand and K_2CO_3 as the base. A series of control experiments indicated that visible light, photocatalyst, copper catalyst, ligand, or base were all indispensable for the successful application of these two transformations.

With the optimal reaction conditions in hand, a broad range of propargyl esters possessing electronically diverse groups were then tested to explore the generality of this photoredox and copper dual-catalyzed reaction (Table 1). Phenylpropargyl esters with a variety of alkyl substituents linked to the ester unit, including linear (**3a**-**3d**), branched (**3e**-**3g**) and cyclic (**3h**-**3k**) substituents, participated well in this transformation, affording the corresponding *S*-alkyl phosphorothioates in 82%−96% yields. Notably, the reaction was easily scaled up to 5 mmol a without significant decrease in the yield of **3a**, and treatment of **3f** with Grignard reagent followed by quenching with H_2SO_4 aqueous solution resulted in the formation of

Table 1. Scope of the *S*-Alkyl Phosphorothioates

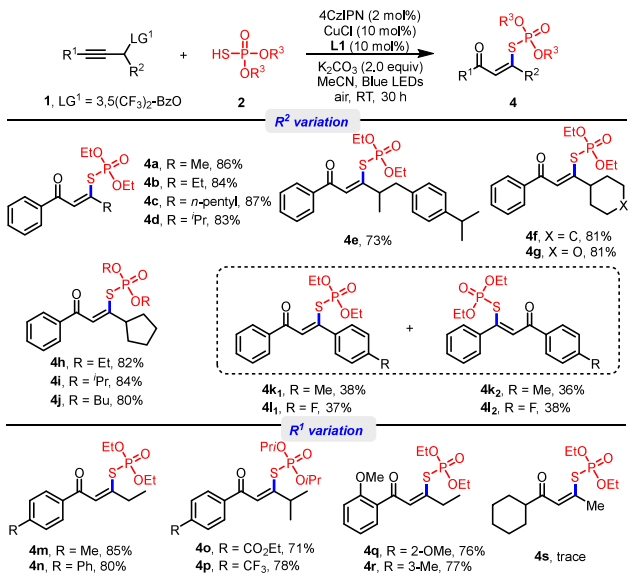


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vinylketone product **3f′**. Functional groups linked to the alkyl substituents, such as the C═C unsaturated bond (**3e**), *oxa*-heterocycle (**3i**) and OTBS (**3l**), were all compatible with this dual catalysis system. In addition to alkyl substituents, aryl substituents linked to the ester unit were also tolerated with the protocol to produce the corresponding products **3m** and **3n** in excellent yields. Thereafter, the variety of alkyne partner tethered to the propargyl ester was investigated. Substrates with electron-donating (Me, Ph, and OMe) or electron-withdrawing (COOEt, CF_3 and F) substituents on the aryl ring smoothly delivered the target products (**3o**-**3x**) in 80%−97% yields. The yields were almost impervious to positions of the substituents on the benzene ring, as demonstrated by **3u**-**3x**. Naphthyl was also tolerated in this protocol (**3y**). In addition to the aryl-containing substrates, substrates with an alkyl (**3z**) or trimethylsilyl (**3aa**) substituent were proven successful for this transformation. The unsaturated bonds, such as alkynyl (**3r**) and alkenyl (**3z**), tethered to alkyne partners were not influenced in this radical transformation.

The generality of the method to synthesize (*Z*)-*S*-vinyl phosphorothioates in air was further explored. As shown in Table 2, phenylpropargyl esters with linear (**4a**-**4c**), branched

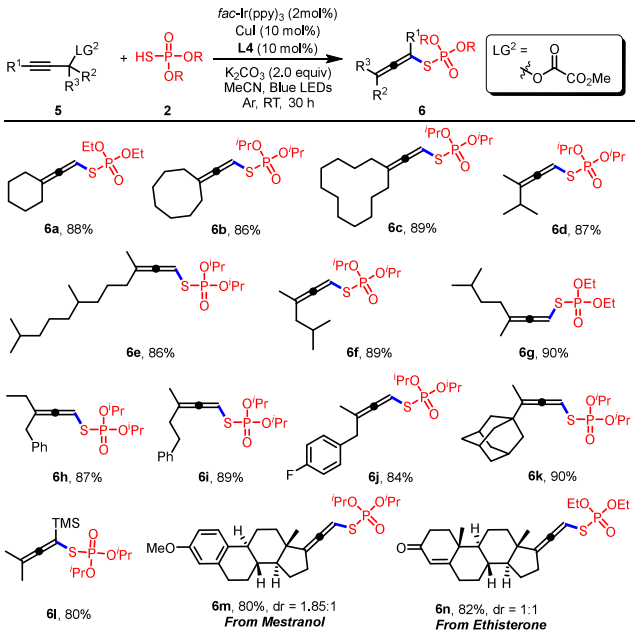
Table 2. Scope of the *S*-Vinyl Phosphorothioates



(**4d**, **4e**) or cyclic (**4f**-**4j**) alkyl substituents linked to the ester unit were all compatible with the protocol to produce structurally diverse *S*-vinyl phosphorothioates in generally good yields. Notably, when phenylpropargyl esters with an aryl substituent linked to the ester unit were used as substrates, mixed products (**4k**, **4l**) were obtain. We suspected that these results might be due to the radical intermediates derived from these substrates possessing little difference in the electronic properties, resulting in poor regioselectivity via the radical shift process. Then a variety of alkyne partners tethered to the propargyl ester were further examined, and the results showed that electron-donating groups (Me, Ph, OMe) and electron-withdrawing groups (COOEt, CF_3) substituted on the *para*-, *meta*- or *ortho*-position of the aryl alkyne moieties were all amenable to deliver the desired products (**4m**-**4r**) in generally good yields. Moreover, other P(O)−SH reagents, including $HSP(O)(O^iPr)_2$ and $HSP(O)(Bu)_2$, were also compatible with this photoredox and copper dual-catalyzed reaction to produce the corresponding products (**4h**, **4i**, **4o**, and **4p**) in satisfactory yields. And an alkylpropargyl ester was also investigated, yet no desired product **4s** was produced.

Then we turned our attention to explore the synthesis of *S*-allenyl phosphorothioates.[4] Fortunately, the target *S*-allenyl phosphorothioates were successfully accessed by using propargylic oxalates (**5**) as the starting material (details see Table S2). Notably, no propargylic isomer was detected under these reaction conditions, demonstrating the excellent regioselectivity of this transformation. As shown in Table 3,

Table 3. Scope of the *S*-Allenyl Phosphorothioates



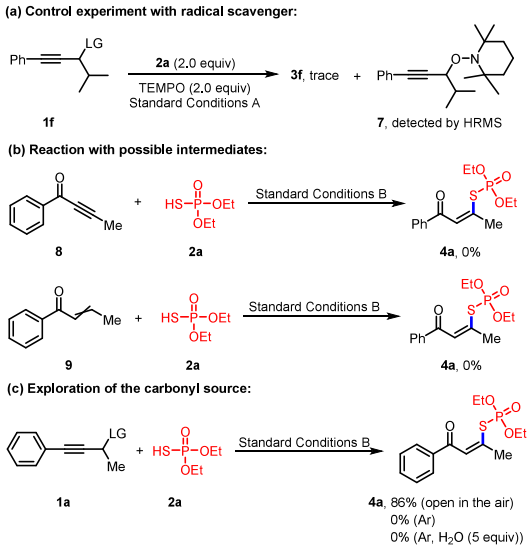
a variety of terminal tertiary propargylic oxalates underwent phosphorothiolation smoothly to produce diverse trisubstituted *S*-allenyl phosphorothioate (**6a**-**6k**, **6m**, **6n**) in good to excellent yields. No significant yield difference among propargylic oxalates with cyclic (**6a**-**6c**) and acyclic (**6d**-**6k**) substituents was observed, demonstrating the good generality of this protocol. Tetrasubstituted *S*-allenyl phosphorothioate (**6l**) was also successfully accessed through this protocol. Furthermore, propargylic oxalates derived from more complex compounds, such as mestranol and ethisterone, served as competent substrates to produce the corresponding products (**6m**, **6n**) in good yields.

Three-component transformations for the synthesis of **3a** and **4a** were also successfully applied using diethyl *H*-phosphonate [$(OEt)_2P(O)H$] and elemental sulfur (S_8) under the standard conditions, yet **6a** was not accessed through this three-component process. An attempt to employ $(OEt)_2P(O)H$ as the coupling partner failed (for details see Scheme S1).

Subsequently, preliminary studies were carried out to provide some insights into the mechanism of this transformation. Initially, when the reaction was performed in the presence of TEMPO (2,2,6,6-tetramethylpiperidinooxy), the target product **3f** was not isolated and TEMPO-trapped product **7** was detected by HRMS, indicating that a radical pathway may be involved in this transformation (Scheme 2a).

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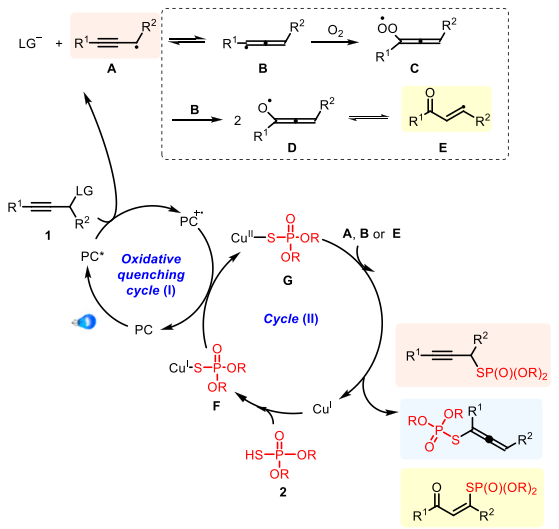
Scheme 2. Mechanistic Studies



No *S*-vinyl phosphorothioate product **4a** was detected when compounds **8** and **9** were used as the starting substrates, thus ruling out these two possible intermediates (Scheme 2b). The desired product **4a** was isolated in an excellent yield under air; on the contrary, no product was detected when the reactions occurring under an Ar atmosphere whether with or without H_2O, which disclosed that the carbonyl source presumably derives from oxygen in the air rather than H_2O (Scheme 2c). Finally, Stern−Volmer luminescence analysis showed the light-activated photocatalyst (4CzIPN*) is more readily quenched by propargyl ester (**1a**) rather than $(EtO)_2P(O)SH$ (**2a**) (for details see Supporting Information).

On the basis of these mechanistic studies and previous reports, a possible mechanism for this photoredox and copper dual-catalytic cycle was proposed in Scheme 3. Upon irradiation with blue light, propargyl derivatives ($E_{1/2}$ [**1a**/**1a**$^{\bullet-}$] = −1.66 V vs SCE in CH_3CN,[9d] $E_{1/2}$ [**5c**/**5c**$^{\bullet-}$] = −1.71 V vs SCE in CH_3CN[10j]) can accept a single electron from the excited state of photocatalyst (PC*) ($E_{1/2}$ [PC$^{\bullet+}$/PC*] = −1.73 V for *fac*-Ir(ppy)$_3$ vs SCE,[12] $E_{1/2}$ [PC$^{\bullet+}$/PC*] = −1.18 V for 4CzIPN vs SCE[13]) leading to the formation of a carboxylic anion (LG^-) and propargyl radical **A** which undergoes a radical shift to generate trisubstituted allenyl radical **B**. Under air, nucleophilic attack of intermediate **B** by oxygen delivers peroxy radical **C**. Subsequently, the interaction between **C** and **B** gives radical **D** and its tautomeric structure alkenyl radical **E**.[11] In the meantime, $(OR)_2P(O)S-Cu(I)$ intermediate **F** is produced through the interaction between Cu(I) species and $(OR)_2P(O)SH$, and then the SET process between **F** and oxidized state of PC$^{\bullet+}$ produces Cu(II) complex **G** with the release of photocatalyst to enclose the photocatalytic cycle. **G** is capable of coupling with radical **A**, **B** or **E** to generate the desired products S-alkyl, S-allenyl or S-vinyl phosphorothioates with the regeneration of Cu(I) species to enclose the Cu-catalytic cycle.

Scheme 3. Proposed Catalytic Cycle



In summary, we have developed a photoredox/copper dual-catalyzed phosphorothiolation between easily accessible [P(O)SH] compounds and propargylic derivatives under mild and operationally simple conditions. Photoinduced formation of a propargyl radical intermediate has been considered as the key step for these transformations. This method provides an efficient protocol for the switchable synthesis of highly valuable *S*-alkyl, *S*-vinyl and *S*-allenyl phosphorothioates from the same set of simple starting materials.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c02636.

Experimental procedures for the synthesis, spectral data, and copies of ^1H, ^{13}C, ^{31}P and ^{19}F NMR spectra of all of the products (PDF)

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Notes

The authors declare no competing financial interest.

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